

Design Calculation Sheet

VBF Case t2_r3 | VBF-T2R3-CALC-01 | Virtual simulation calibre | Generated 2026-08-26

Sheet: 1_inputs

Parameter	Value	Unit	Source
Electrode height × width	0.065 × 1.58	m	Chen2020 dump
Electrode area	0.1027	m2	height × width
Positive electrode thickness	7.56e-05	m	Chen2020 dump
Negative electrode thickness	8.52e-05	m	Chen2020 dump
Separator thickness	1.2e-05	m	Chen2020 dump
Positive current collector thickness (Al)	1.6e-05	m	Chen2020 dump
Negative current collector thickness (Cu)	1.2e-05	m	Chen2020 dump
Positive electrode density	3262	kg/m3	Chen2020 dump
Negative electrode density	1657	kg/m3	Chen2020 dump
Positive porosity	0.335	—	Chen2020 dump
Negative porosity (final)	0.4	—	r6_combo-v5-final_params.json
Positive AM volume fraction	0.665	—	Chen2020 dump
Negative AM volume fraction	0.75	—	Chen2020 dump
c_max positive	63104	mol/m3	Chen2020 dump
c_max negative	33133	mol/m3	Chen2020 dump
Nominal cell capacity	5	Ah	Chen2020 dump
I (1C) = I (discharge) = nominal × 1	5	A	derived

Sheet: 2_capacity_energy

Quantity	Value	Unit	Formula / source
Simulated 1C discharge capacity (DFN, 25 °C)	5.064843296941604	Ah	cell/r7_final_1c_dfn.json:capacity_ah
Discharge energy (DFN time-integral of V·I)	18.45718251889716	Wh	cell/r7_final_energy_dfn.json:energy_wh
Cell energy (kWh)	0.01845718251889716	kWh	Wh ÷ 1000
Midpoint discharge voltage	3.911774309748794	V	cell/r7_final_energy_dfn.json:midpoint_voltage_v
DC resistance (1C)	0.002612451248085001	Ω	cell/r7_final_energy_dfn.json:dcr_ohm

Sheet: 3_energy_density

Quantity	Value	Unit	Formula / source
Cell mass (contract: Σ layer thickness $\times$ (1-porosity) $\times$ density $\times$ area, electrolyte excluded)	41.27970507960001	g	cell/r7_final_energy_dfn.json:mass_kg
Layer masses (pos 16.842 / neg 8.699 / Al 4.437 / Cu 11.042 / sep 0.259)	41.27970507960001	g	r7_final_energy_dfn.json:layer_kg_m2 $\times$ area
Gravimetric energy density	447.1248639811262	Wh/kg	energy_wh $\div$ mass_kg
Cell volume = area $\times$ stack thickness (200.8 μ m)	2.062216e-05	m3	derived
Volumetric energy density	895.0169390062512	Wh/L	energy_wh $\div$ volume_m3 $\times$ 1000
ED incl. electrolyte (estimate 8.016 g)	374.4156	Wh/kg	energy_wh $\div$ (41.280+8.016)g — derived, estimate-caliber electrolyte
Threshold check	PASS (447.12 $\geq$ 327.18)	—	entry-0 criteria, stage2.energy_density_wh_kg.min

Sheet: 4_NP_mass

Quantity	Value	Unit	Formula / source
Positive areal capacity (full-range)	85.027	Ah/m2	$c_max_pos \times AMVF \times L_pos \times F/3600$
Negative areal capacity (full-range)	56.744	Ah/m2	$c_max_neg \times AMVF \times L_neg \times F/3600$
N/P ratio (full-range basis)	0.6674	—	$neg \div pos$ (protocol formula)
Positive full-range cell capacity	8.7323	Ah	$areal \times area$
Usable-window fraction (5.0 Ah design window)	0.5726	—	$nominal \div pos$ full-range
N/P ratio (usable-window basis)	1.1655	—	$neg \div (pos \times fraction)$ — derived variant
Total layer stack thickness	200.8	μm	$75.6+85.2+12+16+12$
Total mass per area	401.9	g/m2	$\Sigma layer_kg_m2$ (r7_final_energy_dfn.json)

Sheet: 5_process

Parameter	Value	Unit	Formula / source
Positive areal density	163.99	g/m2	thickness × (1–porosity) × density
Negative areal density	84.71	g/m2	ditto
Positive compaction density	2.1692	g/cm3	density × (1–porosity) ÷ 1000
Negative compaction density	0.9942	g/cm3	ditto
Electrolyte fill amount	8.016	g/cell	pore volume (6.680 cm3) × 1.2 g/cm3 [E]; fill factor 1.0
Electrolyte fill per area	78.05	g/m2	fill ÷ area
Formation recommendation	0.1C CC to 4.2 V, 25 °C, 2 cycles	—	design-recommended; production-line protocol requires tuning